Geometric and Behavioral Stratification in Transformer Residual Streams

Supplementary Materials · Nelson Guda

Companion to the manuscript (Paper 1). Per-model metric tables and extended output examples moved out of the PDF appendices to keep the arXiv/reading copy compact. Table and example labels are preserved from the appendix numbering they originated in (e.g., Table B.7-2) so cross-references in the manuscript remain traceable. Generated from `Paper_1_v64.md`; Tables S1 and B.7-3 were rebuilt from source data on 2026-08-07, and Table B.6-2 was corrected on 2026-08-11 for the CJK tokenization defect described in §B.6 of the manuscript.

Underlying data files are in the code repository under `data/`; the specific source file is named in each section. Every table in S1 and S2 recomputes from the files named there — the recipes are in the READMEs of `data/classification/`, `data/behavioral_interventions/` and `data/part_g_derived/`.

S1. Part-G layer-angle tracking (single-pass rotation recovery)

Supports the §5.1 / §6.1 statement that a single-pass subspace rotation is partly repaired by later layers. After a one-time rotation at the early intervention layer (12.5–15.0% depth, model-dependent), the rotated subspace's angle relative to its unperturbed orientation is measured at the intervention layer (peak) and at the penultimate layer (L−2). The final layer is excluded because the LM-head projection inflates all residual perturbations uniformly — Mistral 7B, for instance, reads 0.59° at L−2 and 39.48° at the final layer under D-rotation on SpecA. The D-angle under D-rotation is driven above 99° at the intervention site and recovers to below 2° by L−2 in the seven dense non-MoE models; recovery is weaker in the MoE and GPT-OSS architectures. The S-angle under S-rotation behaves the same way from a lower and more variable peak.

Both prompt sets are reported separately below. They differ materially in one place — GPT-OSS 120B's D-angle at L−2 is 4.70° on SpecA and 29.98° on Diverse — so a table that collapsed them would misrepresent the architectural spread.

Table S1 (Part-G per-model angles, degrees). Source: `data/part_g_derived/partG_rotation_recovery_angles.json` in the code repository, extracted from `{model}-Geometry-{set}-part_g_v11_regime_transplant.json`. Values are means over a prompt's scrambles and then over prompts within model. Peak is the angle at the intervention layer itself, not the maximum over layers.

Model	SpecA D peak	SpecA D @ L−2	SpecA S peak	SpecA S @ L−2	Diverse D peak	Diverse D @ L−2	Diverse S peak	Diverse S @ L−2
Mistral 7B	103.5	0.59	50.7	0.89	102.2	1.49	95.6	1.40
Llama 8B	101.1	1.22	82.9	2.65	101.6	1.58	101.8	1.76
Gemma 9B	100.6	0.64	79.8	2.58	99.9	1.06	97.6	1.92
Qwen 14B	100.5	0.64	78.3	0.34	100.5	0.90	95.3	0.43
GPT-OSS 20B	102.0	1.76	80.1	2.71	102.9	3.24	90.0	5.48
Gemma 27B	100.9	0.19	89.5	0.27	104.0	0.30	104.1	1.12
Mixtral 8×7B	103.5	4.30	110.6	2.39	103.3	4.77	106.9	6.37

Model	SpecA D peak	SpecA D @ L−2	SpecA S peak	SpecA S @ L−2	Diverse D peak	Diverse D @ L−2	Diverse S peak	Diverse S @ L−2
Llama 70B	101.2	1.72	89.2	0.99	102.1	0.83	99.4	1.09
Qwen 72B	101.8	0.38	65.6	0.24	102.9	0.49	105.2	1.35
GPT-OSS 120B	102.2	4.70	72.8	6.86	103.3	29.98	106.9	9.29

Cross-model summary, SpecA: D-rotation peak 100.5–103.5°, D-angle at L−2 0.19–1.72° across the seven dense non-MoE models, against 1.76° / 4.30° / 4.70° in GPT-OSS 20B / Mixtral / GPT-OSS 120B. S-rotation reaches a lower and more variable peak (50.7–110.6°, mean 80.0°) and recovers comparably (dense 0.24–2.65°; 2.71° / 2.39° / 6.86°). On Diverse, both rotations peak near or above 100° and the architectural spread at L−2 widens sharply for D (dense 0.30–1.58°; 3.24° / 4.77° / 29.98°) while remaining modest for S (dense 0.43–1.92°; 5.48° / 6.37° / 9.29°). GPT-OSS 120B's extreme residual deviation is therefore specific to D and to the heterogeneous prompt set, and is consistent with the architectural anomalies documented in Appendix C.4.

S2. Per-model persistent-rotation metrics (from Appendix B.7)

Cross-model means/SDs and the interpretive claims these tables support are in the manuscript (§6, Table 3, Appendix B. 7). The per-model breakdowns below are the full backing data.

Table B.7-1b. F-dynamic rotation per-model breakdown, v5 audit corpus, four-category collapsed taxonomy. Failure-rate range across models is reported on the screened denominator (raw rate over all 80 cells per model shown in parentheses where it differs).

Model	N	Excluded	N classified	Failure	Variant	Frame-shift	MSC	Failure rate (screened)
Gemma 9B	80	0	80	80	0	0	0	100.0%
Gemma 27B	80	0	80	80	0	0	0	100.0%
GPT-OSS 20B	80	24	56	56	0	0	0	100.0% (raw 70.0%)
GPT-OSS 120B	80	20	60	60	0	0	0	100.0% (raw 75.0%)
Llama 8B	80	0	80	78	1	1	0	97.5%
Llama 70B	80	0	80	47	24	5	4	58.8%
Mistral 7B	80	0	80	62	11	1	6	77.5%
Mixtral 8×7B	80	0	80	80	0	0	0	100.0%
Qwen 14B	80	0	80	80	0	0	0	100.0%
Qwen 72B	80	0	80	75	5	0	0	93.8%
Cross-model	**800**	**44**	**756**	**698**	**41**	**7**	**10**	**92.3%**

Table B.7-2. D/S behavioral dissociation: per-model divergence metrics, 10 models, SpecB.

Model	D imm%	S imm%	D mean tok	S mean tok	S/D ratio	Random imm%	Cohen's d (D vs Rand)
Gemma 9B	56.2	18.8	1.36	4.09	3.0×	1.6	1.99
Gemma 27B	73.8	15.0	0.49	3.98	8.2×	2.6	1.60

Model	D imm%	S imm%	D mean tok	S mean tok	S/D ratio	Random imm%	Cohen's d (D vs Rand)
GPT-OSS 20B	31.6	13.8	1.71	3.20	1.9×	5.4	1.01
GPT-OSS 120B	28.8	36.2	1.55	3.80	2.5×	3.4	2.09
Llama 8B	79.7	16.5	1.22	6.43	5.3×	1.9	1.93
Llama 70B	79.7	31.9	1.04	12.18	11.7×	3.1	3.06
Mistral 7B	72.5	31.6	1.49	7.22	4.9×	3.8	2.49
Mixtral 8×7B	86.2	61.2	0.65	1.96	3.0×	13.7	1.57
Qwen 14B	66.2	33.8	0.95	5.31	5.6×	9.5	1.96
Qwen 72B	65.0	23.7	1.27	8.16	6.4×	0.0	2.85
Mean ± SD	**64.0 ± 19.8**	**28.2 ± 14.3**	**1.17 ± 0.39**	**5.63 ± 2.99**	**4.8×**	**4.5 ± 4.1**	**2.06 ± 0.58**

GPT-OSS 120B shows a partial anomaly: the mean divergence-token ordering is preserved (D = 1.55 < S = 3.80), but immediate-divergence rates are reversed (S = 36.2% > D = 28.8%). Both GPT-OSS models show attenuated D sensitivity relative to the other architectures (Appendix C.4).

Conventions used in Tables B.7-2 and B.7-3. *Imm-div %* is the share of prompts whose first divergence occurs at token 0, taken over the prompts for which a divergence token is defined — prompts whose intervened output is text-identical to the baseline are excluded from the denominator rather than counted as non-immediate. *Mean first-div token* is the mean over the same prompts. *Cohen's d* is the pooled-SD standardized difference between the condition's and the Random control's first-divergence-token distributions over all 80 prompts, with identical-output prompts assigned the full 64-token generation length. Both tables are recomputed from `data/behavioral_interventions/{model}-Continuation-SpecB-{condition}.json` in the code repository.

Table B.7-3. F-topK per-model divergence metrics, SpecB, 10 instruction-tuned models.

Model	Imm-div %	Mean first-div token	Cohen's d (F-topK vs Random)
Gemma 9B	15.0	4.25	1.79
Gemma 27B	30.0	1.73	1.49
GPT-OSS 20B	7.6	5.99	0.69
GPT-OSS 120B	13.9	8.80	1.34
Llama 8B	13.8	7.84	1.60
Llama 70B	27.5	6.10	2.73
Mistral 7B	21.1	11.59	1.34
Mixtral 8×7B	53.8	2.71	1.39
Qwen 14B	25.0	7.04	1.50
Qwen 72B	23.8	9.14	2.11
Mean ± SD	**23.1 ± 12.9**	**6.52 ± 3.03**	**1.60 ± 0.54**

F-topK's immediate-divergence rate ranges from 7.6% (GPT-OSS 20B) to 53.8% (Mixtral 8×7B); mean first-divergence token ranges from 1.73 (Gemma 27B) to 11.59 (Mistral 7B). This within-condition variability tracks the corresponding S-scramble values closely across models, supporting the S-like-regime claim. The main model-specific nuance is Mixtral

8×7B, where F-topK Frame-shift reaches 15.0% (comparable to S-scramble's 16.2% on the same model and above the cross-model F-topK mean of 6.5%), suggesting a soft overlap in some MoE architectures between the directions used by D and the highest-variance directions inside F.

Table B.7-4. Frame-shift rate per model and condition under the collapsed taxonomy.

Model	D-scramble	S-scramble	F-topK	Random
Qwen 14B	61.3%	8.8%	7.5%	1.2%
Mixtral 8×7B	50.0%	16.2%	15.0%	6.2%
Llama 8B	48.8%	2.5%	5.0%	2.5%
Qwen 72B	48.8%	10.0%	7.5%	0.0%
Mistral 7B	36.2%	8.8%	3.8%	0.0%
Llama 70B	31.2%	6.2%	7.5%	0.0%
Gemma 9B	25.0%	7.5%	2.5%	1.2%
GPT-OSS 20B	2.5%	1.2%	1.2%	1.2%
GPT-OSS 120B	1.2%	3.8%	2.5%	0.0%
Gemma 27B	0.0%	10.0%	7.5%	1.2%

Seven of ten models show D-scramble Frame-shift rates between 25.0% and 61.3%. The three exceptions are Gemma 27B (D's effect concentrates in Variant and Failure) and the GPT-OSS family (baseline degeneracy and D-attenuation route many cells into Failure).

S3. F-structure interventions across depth (from Appendix B.5)

The aggregate result and Table B.5-1 are in the manuscript (Appendix B.5). Per-model identical-output rates at 70% depth:

Table B.5-2. Per-model identical-output rates at 70% depth.

Model	F-mix ident. %	F-attenuate ident. %	F-mix < F-attenuate?
Mistral 7B	7.1	42.9	✓
Llama 8B	11.9	60.7	✓
Gemma 9B	27.4	60.7	✓
Qwen 14B	7.1	42.9	✓
GPT-OSS 20B	1.2	33.3	✓
Gemma 27B	1.2	46.4	✓
Mixtral 8×7B	1.2	31.0	✓
Llama 70B	22.6	71.4	✓

Model	F-mix ident. %	F-attenuate ident. %	F-mix < F-attenuate?
Qwen 72B	8.3	50.0	✓
GPT-OSS 120B	8.3	34.5	✓
Mean ± SD	**9.6 ± 8.9**	**47.4 ± 13.4**	**10/10**

Data source: `analysis/f_intervention__behavioral__d70_residual/Revised Analysis/group_e_depth_70_consolidated_results.json` (70% depth) and `early_layer_F_intervention_results_revised.json` (12% depth).

S4. F-intervention failure-mode distribution at 70% depth (from Appendix B.6)

The reclassification narrative and the ~12% depth table (B.6-1) are in the manuscript. The 70% depth replica:

Table B.6-2. Failure-mode distribution across F-structure-intervention continuations at ~70% depth (reclassified).

Outcome	F-Mix	F-Attenuate
Identical	9.6% (81)	47.4% (398)
Minor reframe	68.6% (576)	44.8% (376)
Mode/stance change	17.0% (143)	6.7% (56)
Topic drift	1.8% (15)	0.7% (6)
Failure mode	3.0% (25)	0.5% (4)
Total N	**840**	**840**

Corrected for the CJK tokenization defect described in §B.6 of the manuscript. 71 trials at this depth move out of the failure category (55 to minor reframe, 16 to mode/stance change); none moves into it, and the identical and topic-drift counts are unchanged. The superseded values were 64.6% (543) / 15.8% (133) / 8.1% (68) for F-Mix and 42.1% (354) / 6.0% (50) / 3.8% (32) for F-Attenuate.

S5. Base vs. instruct per-model detail (from Appendix C)

Base-model dimensionality (Tables C.1–C.2) and the convergence summary (Table 2, Figure 5) are in the manuscript. Per-pair D-angle separation and per-model spectral compression:

Table C.3. D-angle separation improvement under instruction tuning (SpecA, 8 matched pairs).

Model	Base Gap (°)	Instruct Gap (°)	Δ	Factor
Llama 8B	17.8	38.2	+20.4	2.1×
Llama 70B	24.8	45.8	+21.0	1.8×
Qwen 14B	20.7	33.4	+12.7	1.6×

Model	Base Gap (°)	Instruct Gap (°)	Δ	Factor
Qwen 72B	17.8	29.2	+11.4	1.6×
Gemma 9B	26.9	40.4	+13.5	1.5×
Gemma 27B	7.8	39.7	+31.9	5.1×
Mistral 7B	23.6	40.7	+17.1	1.7×
Mixtral 8×7B	18.0	27.1	+9.1	1.5×

Table C.4. Spectral dimensionality k_D changes under instruction tuning (base → instruct).

Model	SpecA B→I	SpecB B→I
Llama 8B	8→7	21→28
Llama 70B	13→6	8→38
Qwen 14B	10→6	13→31
Qwen 72B	11→4	7→28
Gemma 9B	9→5	20→19
Gemma 27B	2→5	2→12
Mistral 7B	18→5	19→27
Mixtral 8×7B	12→8	17→59

S6. Extended intervention output examples (from Appendix D)

The manuscript retains six representative examples. These three additional examples illustrate the floor of the F-mix Minor reframe category, a typical D-scramble Frame-shift, and the minority S-scramble Frame-shift case. All interventions are at ~70% depth during greedy autoregressive generation (64 tokens).

S6.1 F-mix: minor reframe (the mildest outcome)

Model: Gemma 9B Instruct | **First divergence token:** 5

This example illustrates the minimal floor of the F-mix Minor reframe category: the output differs from baseline by a single punctuation mark and continues identically thereafter. The Minor reframe category spans from this floor through to substantive content rewrites and accounts for ~68.6% of F-mix outcomes overall.

Prompt: *Machine learning models can fail in production for reasons that are not apparent during training. The most common failure modes include*

Baseline:

> You're absolutely right! Machine learning models can be tricky beasts, often behaving unexpectedly in real-world scenarios despite performing well during training. [continuation truncated; identical to F-mix from this point]

F-mix:

> You're absolutely right. Machine learning models can be tricky beasts, often behaving unexpectedly in real-world scenarios despite performing well during training. [continuation truncated; identical to baseline from this point]

The only difference is "right!" → "right." — an exclamation mark replaced by a period. The remaining ~50 tokens of each continuation are token-for-token identical. Destroying the directional structure of >99% of the residual stream changed a single punctuation mark.

S6.2 D-scramble: framing shift to writing-coach (a typical D-scramble Frame-shift)

Model: Llama 8B Instruct | **First divergence token:** 0 | **v4 subtype:** MSC (mode/stance change)

This example illustrates a typical D-scramble Frame-shift: the baseline produces a six-word narrative twist; D-scramble reframes the prompt as a writing exercise and generates first-person backstory. Frame-shifts of this kind account for 42.3% of D-scramble outputs (Table 3).

Prompt: *As the family's oldest son, he was expected to take over the business.*

Baseline:

> ...but he had other plans.

D-scramble:

> That's a great starting point. Here's a potential story based on that premise:
>
> I've always known that I was destined for greatness. As the oldest son in a family of modest means, I was expected to take over the family business. My parents had built a small but successful convenience store, and it was my...

The communicative mode flips from in-narrative continuation to writing-coach reframe ("Here's a potential story based on that premise"). The prompt's narrative invitation is preserved as content, but the model now treats it as a writing assignment to discuss rather than a story to continue.

S6.3 S-scramble: rare framing shift (the minority 8.8% case)

Model: Gemma 9B Instruct | **First divergence token:** 0 | **v4 subtype:** MSC

S-scramble usually preserves the baseline's framing and only redirects content (77.1% Variant). This example illustrates the rare 8.8% Frame-shift case: the baseline reads the prompt as personal disclosure and asks empathetic follow-up questions, while S-scramble reframes the same prompt as a story prompt and offers writing-coach prompts for elaboration.

Prompt: *I started packing my bags for the trip I'd been planning for years.*

Baseline:

> That's exciting!
>
> Tell me more about this trip. Where are you going? What have you been looking forward to the most?

S-scramble:

> This is a great start to a story!
>
> I'm curious to know more.
>
> - **Where are you going on this trip?** Is it a relaxing beach vacation, an adventurous backpacking expedition, or something else entirely?
> - **Why have you been planning this trip for so long?**

Both responses ask questions and use a friendly tone, but the baseline addresses the user's experience while S-scramble treats the same sentence as a writing prompt. The mode shift is from `empathetic_listener` to `writing_coach` — present in only 8.8% of S-scramble outputs but qualitatively similar to D-scramble's modal Frame-shift profile.